One-Dimensional Interior Models of Europa

Isabelle Joosten
Student number 4998480
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1 Introduction

Europa is one of Jupiter's largest moons. It is a scientifically interesting planetary body, as it is commonly thought to have a subsurface liquid water ocean that may be capable of harbouring life. In this report, various models of Europa's interior found in literature are analysed and compared. Based on these, three new models are constructed, with the aim of characterising Europa's global interior.

2 Literature Study

Several existing models of Europa's interior were consulted for the purposes of this assignment. Jin and Ji[1] published an article in which they outline the creation of 3 models of Europa, the difference between the models being the assumed composition of the core: Fe, FeS, or $Fe_{0.8}(FeS)_{0.2}$. Aside from the core, Jin and Ji define two more layers in each of their models: an olivine mantle and an outer shell made of water. To generate interior profiles of Europa, they choose a fractional mass for each of the 3 layers as well as a centre pressure. The program they developed performs an integration from the centre to the surface in order to solve for the surface pressure, logically must be 0. Based on the calculated surface pressure, the program then adjusts the centre pressure before restarting the integration, repeating this sequence until the calculated surface pressure is 0.

Another existing model found in literature is one published by Cammarano et al.[2]. The aim of their research was to find the seismic velocities associated with different possible interior of Europa, so their model determines the density profile based on temperature and composition. Like Jin and Ji, Cammarano et al. assume an FeS core and a water shell, but unlike Jin and Ji, they consider two different mantle compositions: pyrolitic and chondritic.

Kuskov and Kronrod[3] base their model on the observed mass, bulk density, and moment of inertia of Europa. Unlike the two models perviously discussed in this section, in this model the mantle is divided into 3 layers, leading to 5 layers in total.

Finally, Howell[4] focuses on Europa's icy outer shell, and attempts to determine the thickness of this shell based on its steady-state heat flux. This model differs from the others in that it does not provide a profile of Europa's full interior, but rather provides a detailed profile of the ice crust based on assumed values for the rest of the interior.

Because all models found use a different approach and were created with a different goal in mind, a direct comparison is difficult to make. Nevertheless, there are a few notable trends among the articles. There seems to be a general consensus among researchers that Europa consists of an iron-rich core, a silicate or chondritic mantle, and an ice/water outer shell[1]. Numerically, there is some variation between the outputs of each model, part of which can be explained by the fact that the models use different theoretical values as inputs, most of which are, themselves, obtained through theoretical modeling. A summary of relevant values is given in Table 1. As each of these models has undergone some form of verification and validation before being published, these values, or averages thereof, will be used as reference values for the remainder of this report.

3 General Modeling Approach

For this report, three one-dimensional models of Europa's interior were created. Models I and II are written from scratch in Python, with model II being an extension of the simpler model I. Model III is written using the BurnMan Python module, and the setup of this model will be described in Section 7.

An overview of the input and output parameters used in model I is given in Table 3. The approximate density of the core, mantle, and outer shell are taken from Jin and Ji[1]. The simulation starts by upward integrating the mass for each step in radius and calculating the gravitational acceleration at each of those radii, according to Equations 1 and 2, respectively. Then, the simulation integrates downward according to Equation 3 to find the pressure at each radius step.

$$\frac{dM}{dr} = 4\pi\rho r^2 \tag{1}$$

[1]It should be noted, though, that this is by no means definitively proven. See for example Trinh et al.[5], in which the existence of Europa's metallic core is questioned based on reanalysis of Galileo data.

Table 1: An overview of the models analysed for this research.

	Jin & Ji [1]	**Cammarano et al. [2]**	**Kuskov & Kronrod [3]**	**Howell [4]**
Publication year	2012	2006	2005	2021
Total radius [km]	1563	-	1565	1561
Core radius [km]	777	~600	880	630
Shell thickness [km]	167	~100	105-160	127
Moment of inertia [-]	0.330	0.346	0.346	0.359
Total mass [kg]	$4.80*10^{22}$	$4.8*10^{22}$	$47.99*10^{21}$	$4.80*10^{22}$
Centre pressure [GPa]	~5.5	-	~4.8	-

$$g(r) = \frac{GM(r)}{r^2} \tag{2}$$

$$\frac{dp}{dr} = -\rho g(r) \tag{3}$$

Once this is done, the dimensionless moment of inertia is calculated, using Equation 4.

$$\bar{I} = \frac{8\pi}{3MR^2}\int_0^R \rho(r) r^4 dr \tag{4}$$

This is then compared to the observed dimensionless moment of inertia. If the calculated moment of inertia is higher than the observed value, the layer heights are adjusted such that the mass of the moon is more concentrated in the centre, by decreasing either the core radius or the mantle thickness. If it is lower than the observed value, the layer heights are increased. The total calculated mass and mean density are also compared with the observed values, and the layer heights are adjusted accordingly.

To avoid the model getting stuck in an infinite loop where it decreases and then immediately increases a layer height by the same amount between two iterations, the layer being adjusted is chosen at random, where both layer heights have an equal chance of being picked. The amount by which the height is adjusted is equal to the 1 km step sizes times a scaling factor. This scaling factor, in turn, is the product of the residual mass, moment of inertia, or mean density, divided by the uncertainty of the observation, multiplied by a random number between 0.7 and 1.3. The scaling factor allows the program to adjust the layer heights in larger steps when the residual is very large, significantly speeding up the process, while the random multiplier again ensures the program does not get stuck in a loop.

Once the layer heights have been adjusted, the simulation starts over again, and it continues in this loop until the calculated mass, moment of inertia, and mean density are within the uncertainty range of the observed value. The observed values can be found in Table 2. Model II works in the same way, except the density values are now dependent on temperature and pressure.

Table 2: The observed values used for comparison of with the simulations' output values.

	Value	**Uncertainty**	**Source**
Mass	$479.7 * 10^{20}$ kg	$1.5 * 10^{20}$ kg	Yoder, 1995
Moment of inertia factor	0.346	0.005	Schubert et al
Mean density	3.0130 g/cm^3	0.0017 g/cm^3	NASA JPL SSD

4 Results, Verification and Validation of Model I

To construct model I, the following assumptions were made:

- Europa consists of three layers: an FeS core, a pyrolitic mantle, and an outer shell of water and ice;
- Because it is expected to be relatively thin compared to the rest of the water shell, the ice crust is neglected and the outer shell is assumed to be 100% liquid water[1];
- The density of each layer is homogeneous and is 5500, 3300, and 1000 kg/m^3 for the core, mantle, and outer shell, respectively. These values were estimated from the graphs published by Jin and Ji[1];

The initial core radius and outer shell thickness are chosen somewhat arbitrarily: based on the values found in Table 1, the core-mantle boundary is initially set to 700 km from the centre, and the mantle-shell boundary is set to 1400 km from the centre, which gives an outer shell thickness of 165 km. The step size with which to vary the layer boundaries is chosen to be 1 km. Various step sizes were tested, and 1 km was found to be a good compromise between

Table 3: An overview of the input and output parameters for model I.

Inputs	**Outputs**
Observed radius	Mass
Observed mass	Gravity
Observed moment of inertia	Pressure
Observed mean density	Moment of inertia
Core density	Core radius
Mantle density	Shell thickness
Shell density	Mean density
Initial core radius	
Initial shell thickness	

accuracy and calculation time: it is small enough to allow the program to converge to a layer distribution that fits the observed values to within their margins of uncertainty, yet large enough to allow this convergence to occur within less than a minute, allowing for many tests in quick succession.

Running the program with these inputs results in the values shown in Table 5. Furthermore, the model predicts the pressure at the centre of Europa to be 5.24 GPa, which roughly agrees with the values in Table 1. The gravitational acceleration at the surface is found to be 1.31 m/s^2, matching the value given by Cammarano et al.[2]. The core radius and shell thickness found are 719 and 122 km, respectively, roughly resembling the models in Table 1, though not quite matching any of them. It takes the program 1291 iterations to converge to these values.

To verify the model, a few tests were done. The sensitivity of the code was checked by running the model multiple times, each time with slightly different values for the initial core- and mantle boundaries. To avoid the randomness in the layer height adjustment skewing the result, each combination of boundaries is repeated ten times. The added benefit of running the model in this way is that the resulting layer heights of all runs can be averaged, allowing for conclusions to be drawn with more confidence than if the conclusions were based on a single run.

This verification program was run using a range of initial core boundaries from 695 to 705 km and a range of initial mantle boundaries from 1395 to 1405 km, with an interval of 1 km. That means the standard deviation is 3.164 km for both sets of initial values. Some statistical measures of the output values are shown in Table 4. Notably, the standard deviations for both values are higher than 3.164 km, indicating that the spread of the output values is wider than that of the input values, which can also be seen in the minimum and maximum values. In particular, the computed core boundaries are rather varied, and the large difference between the mean and median values implies there may be outliers skewing the result.

While it is possible that this result indicates an error in the simulation, it is also possible that it works as intended,

Table 4: Results of the sensitivity test for model 1. All values are given in kilometres.

	Core boundary	**Mantle boundary**
Standard deviation	25.25	5.154
Mean	698.5	1442
Median	708.0	1440
Range	636.0 - 724.9	1435 - 1455

and that there simply are multiple combinations of layer heights that satisfy the conditions set by the observations. More testing is needed to draw proper conclusions. However, for the purpose of this research, the program is considered to be working well enough to continue with further steps.

The model was validated by changing all the inputs to Earth's characteristics, as these are known with more certainty than Europa's characteristics[2]. The model decently accurately predicts the gravitational acceleration at Earth's surface (9.804 m/s^2) and generates a pressure curve that looks very similar to a plot based on the Preliminary Reference Earth Model (PREM) published by Alfè et al.[?], although the maximum pressure predicted by the model (351.9 GPa) is a bit lower than that in the PREM (363.9 GPa)[?]. It also quite accurately predicted the radius of Earth's core at 3497 km, compared to the PREM's 3480 km for the inner and outer core combined[?]. Interestingly enough, however, it seems to favour an unrealistically thick crust regardless of the initial conditions set. This could be a consequence of the somewhat oversimplified criteria for adjusting the core radius and crust thickness, or of the assumption that the density of each layer is homogeneous. Such an assumption should more significantly affect a model of Earth than of Europa simply because Earth is so much more massive. It can be concluded that although it is a simplified model, it is accurate and realistic enough for the purpose of this report.

5 Thermal Effects

To construct model II, model I is extended to include the effects of temperature on the density of each of Europa's layers. To this end, three thermal profiles are constructed, based on three centre temperature values found in literature: a minimum temperature, a maximum temperature, and a mean temperature.

Information about the core temperature of Europa is not easy to find. When it is mentioned in literature, it appears to be mostly speculative values, which makes sense, given

[2]The thickness and density of each layer was taken from `http://hyperphysics.phy-astr.gsu.edu/hbase/Geophys/earthstruct.html`.

Table 5: The results of model 1 in comparison to observed values.

	Result	**Difference with observed value**	**Fraction of uncertainty**
Mass	4.788*10^{22} kg	9.005*10^{19} kg	0.600
Moment of inertia	0.341	0.0046	0.9233
Mean density	3011.8 kg/m^3	1.2 kg/m^3	0.678

the limited information available about Europa in general. Cammarano et al.[2], for example, argue that for the core to be solid, its temperature must be below 1900 K for a pure iron core, and below 1400 for an FeS core. Travis et al.[7] propose a core temperature of 1750 K based on their simulations of the thermal history of Europa. Kuskov and Kronrod[8] found a temperature of 1293 degrees Celsius, or 1566 K, at the core-mantle boundary using one of their simulations. Finally, Prentice[9] modeled Europa with a core temperature of 1073 K.

Table 6: The centre temperature values chosen as basis of the three thermal profiles for Europa.

Min	1073 K
Mean	1447 K
Max	1750 K

Based on these values, three core temperatures were selected, as shown in Table 6. Using a linear temperature profile ranging from the core temperature to the surface temperature, chosen to be 104 K[4], a density profile can be constructed using the following equation:

$$\rho(T) = \rho_0(1 - \alpha_T \Delta T + \frac{1}{K}\Delta p) \tag{5}$$

In the above equation, α_T represents the thermal expansivity of the material, and K represents the bulk modulus of the material. ρ_0 represents a reference density. The values used in model II to calculate the density as a function of the temperature are provided in Table 7.

Table 7: The values used to determine the density at each point in the profile.

	Core	**Mantle**	**Shell**
Material	Iron sulfide	Pyrolitic	Water
Density [kg/m^3]	5500	3300	1000
Expansivity [K^{-1}]	87*10^{-6} [10]	40*10^{-6} [11]	-4.93*10^{-8} [12]
Bulk modulus [GPa]	75.2 [13]	130 [14]	10.56 [15]

Equation 5 is implemented into model I by using it to update the density profile after each iteration. Because Equation 5 requires the pressure as an input, and the density profile is required to calculate the pressure profile, the first iteration is done using the same homogeneous density profile used in Model I. Aside from these things, the model works in the same way as model I.

Table 8: The results of model II for the minimum core temperature, mean core temperature, and maximum core temperature.

	Min	**Mean**	**Max**
Mass [10^{22} kg]	0.212	0.941%	0.956%
Centre pressure [GPa]	5.201	4.921	4.921
Gravity at surface [m/s^2]	1.315	1.295	1.295
Core radius [km]	721.5	666	677
Shell thickness [km]	122.9	123	120
Moment of inertia [-]	0.342	0.0350%	0.0256%
Mean density [kg/m^3]	3013	x	x
Iterations	140	66	77

6 Results, Verification and Validation of Model II

The results of model II are summarised in Table 8. To validate model II, the resulting graphs were qualitatively compared with graphs found in literature. Comparing model II to the model created by Jin and Ji[1] shows that the shape of the plot is very similar to that plotted by Jin and Ji. It is interesting to note that although the shapes of the plots are quite similar, the centre pressure obtained by model II is lower than any of Jin and Ji's models by almost 0.5 GPa. Comparing model II to the values given in Table 1, all values roughly match those found in literature.

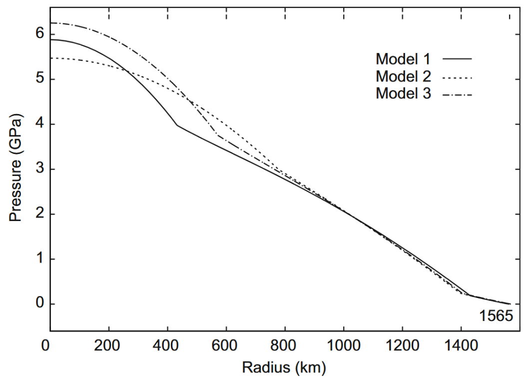

Figure 1: The interior pressure of Europa as determined by Jin and Ji[1].

7 Results and Discussion of Model III

Model III makes use of the Burnman module available for Python and was created by modifying the file "tutorial_03_layers_and_planets.ipynb" included in the Burnman installation. The Burnman module includes a mineral library, but it is somewhat limited. Thus, for model III, slightly different materials were chosen than for models I and II: the core is assumed to be pure iron rather than iron(II)sulfide, the mantle is assumed to consist of Olivine, and the outer shell is assumed to be entirely liquid water. For each layer, the same linear temperature variation is used as in model II, and the pressure mode is set to "self-sufficient". The initial values for pressure and gravity are taken from the results of model II (at the mean core temperature), and so are the layer thicknesses. The results of this model, as well as a comparison to models I and II, are shown in Figure 2. An overview of some key parameters computed by model III are shown in Table 9.

Table 9: The results of model III for the minimum core temperature, mean core temperature, and maximum core temperature.

	Min	**Mean**	**Max**
Mass deviation	16.23%	16.58%	17.07%
MOI deviation	0.0286%	0.0934%	0.0282%
Iterations	5	5	5

8 Comparison Between Models

When viewing Figure 2, one difference between the models immediately stands out. While models I and II match almost perfectly, model III consistently computes higher values for density, pressure, and gravity. One way this could be explained is the difference in assumed layer compositions. In the core, the figure shows the density found in model III is over 25% higher than that for models I and II. This makes sense when considering that the core is assumed to consist of iron (II) sulfide (FeS) in models I and II and of pure iron (Fe) in model III. By comparing the atomic weights of iron and sulfur it logically follows that pure iron would have a higher density. Since gravity and pressure are both functions of density, this difference, along with the smaller but still significant differences in mantle and shell density, can account for the differences in gravity and pressure found.

9 Conclusion

In this report, four interior models from literature were analysed and compared to each other. Furthermore, two new models were constructed and compared to both each other and the models found in literature. These two models have shown to provide a decent first estimate for the interior profile of Europa, and they roughly agree with the models found in literature. However, there is a lot of uncertainty in the input values taken from literature. Future missions to Europa could provide more accurate data that can be used to construct more accurate models. The models themselves are quite simplified, and neglect many factors necessary to create a realistic model. In future research, these models could perhaps be expanded to include more than 3 layers, and thermal effects such as tidal heating or radiogenic heating.

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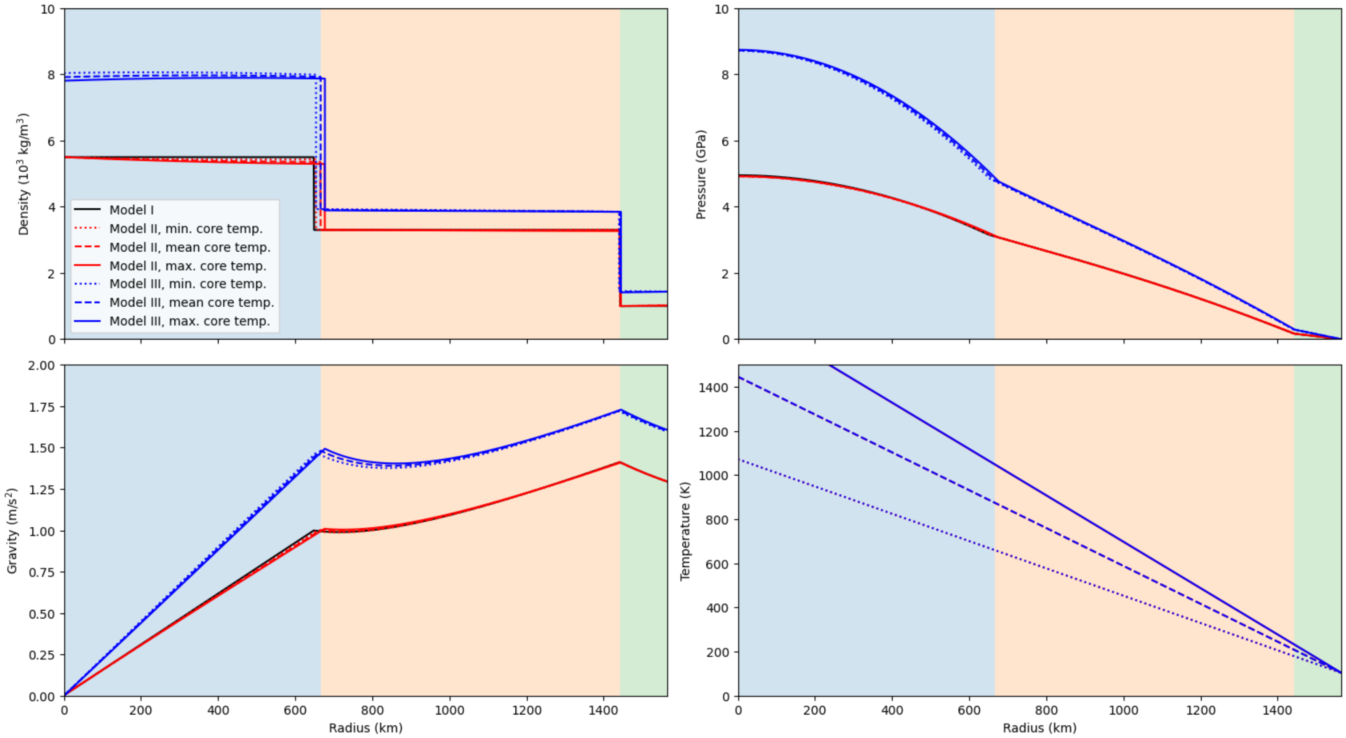

Figure 2: The density, pressure, gravity, and temperature profiles of Europa as calculated by the 3 different models.

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